Chapter 9. Fourier Series

Orthonormal systems, pointwise convergence, Fejér's theorem, Parseval's identity, smoothness and decay

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9.1 Fourier coefficients

Motivation. If a periodic function is a trigonometric series, its coefficients must be given by integrals against $\cos nx$ and $\sin nx$. We study these coefficients for arbitrary integrable functions: they are the best L^2 approximation and decay to zero.

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n} \hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\| x - \sum_{n=-N}^{N} c_n\phi_n \Big\| \le \Big\| x - \sum_{n=-N}^{N} z_n\phi_n \Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}} |c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x-s, \phi_n\rangle = 0$ for $|n| \le N$, so $x - s \perp s - t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum |c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form,

$$s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n} (a_k \cos kx + b_k \sin kx)$$

with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum (-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}} \frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k \text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Motivation. Does the Fourier series of f converge back to f? Not always, but it does at points where f is reasonably smooth. The Dirichlet kernel reduces the question to an integral, and the Riemann–Lebesgue lemma handles it.

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty} \int_I f(t)\sin(\alpha t+\beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t+\beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I |f-s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ■

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the **Dirichlet kernel**:

$$s_n(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t) = \sum_{|k|\le n} e^{ikt} = \frac{\sin(n+\frac{1}{2})t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n = 1.$$

Since D_n is even, $s_n(f;x) = \frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0 < \delta < \pi$ and $\int_{-\pi}^{\pi}|f| < \infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac{1}{2})t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)} - \frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)} - \frac{2}{t})\sin(n+\frac{1}{2})t\,dt \to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac{1}{2})t}{t}$ also tends to 0. ■

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt = \lim\int_0^{\lambda\delta}\frac{\sin u}{u}du = \frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt < \infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Write $g(t) = g(0+) + (g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ■

Theorem 9.6 (9.2.4, Dini's test)

Let $f \in \mathcal{R}^1[-\pi,\pi]$ and $x \in [-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt < \infty$. Then $s_n(f;x) \to \frac{1}{2}[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$, with $\lambda = n+\frac{1}{2}$. ■

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)| \le Mt$ for small $|t|$, then $s_n(f;x) \to f(x)$.

Proof. The integrand in (b) is at most $2M$. ■

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x) \to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)| \le (|f'(x)|+1)|t|$ for small t. ■

Lemma 9.9 (9.2.7)

If $g:[0,\delta]\to\mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^\delta g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. By Jordan decomposition we may assume g increasing, and subtracting a constant, $g(0+)=0$. Given ϵ, choose $h\in(0,\delta)$ with $g(h)<\epsilon$. On $[h,\delta]$, $g(t)/t$ is integrable, so that part tends to 0 by Theorem 9.3. On $(0,h]$, by the second mean value theorem (Corollary 5.25 of Volume I, applied on $[\eta,h]$ and letting $\eta\to 0$), $\int_0^h g(t)\frac{\sin\lambda t}{t}dt = g(h-)\int_c^h \frac{\sin\lambda t}{t}dt$ for some $c\in[0,h]$. Since $|\int_a^b \frac{\sin u}{u}du|\le K$ for all $0\le a<b$ (a bounded quantity, as $\int_0^\infty \frac{\sin u}{u}$ converges), this part is at most ϵK. ■

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is of bounded variation on some interval $[x-\delta,x+\delta]$, then $s_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$.

Proof. $g(t)=\frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0,\delta]$ with $g(0+)=\frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ■

Example 9.2

For $f(x)=x$ on $(-\pi,\pi)$ the series converges to x inside and to 0 at $\pm\pi$; at $x=\pi/2$ it gives $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots$. Near the jump the partial sums overshoot by about 9% of the jump (Gibbs phenomenon). There exist continuous functions whose Fourier series diverge at a point (du Bois-Reymond).

9.3 Fejér's theorem

Motivation. Even when a Fourier series diverges, averaging its partial sums converges uniformly to any continuous periodic function. This shows that a continuous function is determined by its Fourier coefficients and gives another proof of Weierstrass approximation.

The **Cesàro means** are $\sigma_n(f;x)=\frac{1}{n+1}\sum_{k=0}^n s_k(f;x)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t)=\frac{1}{n+1}\sum_{k=0}^{n}D_k(t)=\frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n\ge 0$, $\frac{1}{2\pi}\int K_n=1$, and $K_n(t)\le\frac{1}{(n+1)\sin^2(\delta/2)}\to 0$ uniformly on $\delta\le|t|\le\pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t\cdot\sin\frac{t}{2}=\frac{1}{2}\sum(\cos kt-\cos(k+1)t)=\sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f\in\mathcal{R}^1[-\pi,\pi]$ and $f(x\pm)$ exist, then $\sigma_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f)\to f$ uniformly.

Proof. Since K_n is even, $\sigma_n(f;x) - s = \frac{1}{2\pi}\int_0^\pi [f(x+t)+f(x-t)-2s]K_n(t)\,dt$ with $s = \frac{1}{2}[f(x+)+f(x-)]$. Given ϵ, choose δ with $|f(x+t)+f(x-t)-2s| < \epsilon$ for $0 < t < \delta$. The integral over $(0,\delta)$ is at most $\epsilon \cdot \frac{1}{2\pi}\int_0^\pi K_n \le \epsilon/2$, and that over $[\delta,\pi]$ is at most $\frac{1}{(n+1)\sin^2(\delta/2)} \cdot \frac{1}{2\pi}\int |f(x+t)+f(x-t)-2s| \to 0$. For continuous periodic f, δ can be chosen uniformly in x and f is bounded, giving uniform convergence. ■

So every continuous periodic function is a uniform limit of trigonometric polynomials (the trigonometric form of Weierstrass's theorem), and a continuous periodic function is determined by its Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \ge N$. For $n > N$, $|\sigma_n| \le \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Given ϵ, choose a continuous g vanishing near $\pm\pi$ (so continuous as a periodic function) with $\|f-g\|_2 < \epsilon$ (Theorem 7.14 and Lemma 7.13), and by Theorem 9.11 a trigonometric polynomial $T = \sigma_m(g)$ of degree m with $\|g - T\|_{\sup} < \epsilon$, so $\|g-T\|_2 < \epsilon$. For $n \ge m$, Theorem 9.1 gives $\|f - s_n(f)\|_2 \le \|f - T\|_2 < 2\epsilon$. ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a+2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I |f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\le n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k \|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = 2\sum_{n\ge1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Motivation. Fourier series can always be integrated term by term, and differentiated term by term under smoothness assumptions. These rules make Fourier series a practical tool for differential equations.

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum \hat f(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat f(n) \int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $|\int_a^x (f - s_n(f))| \le \sqrt{2\pi}\Big(\int_I |f - s_n(f)|^2\Big)^{1/2} \to 0$ by Corollary 9.14. ∎

Note that the integrated series converges whether or not the Fourier series of f converges. If f is periodic and C^1, integration by parts gives $\widehat{f'}(n) = in\hat f(n)$; since $\sum |n\hat f(n)|^2 < \infty$, Cauchy–Schwarz gives $\sum |\hat f(n)| < \infty$, so the Fourier series converges uniformly (to f, by Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat f(n) = a(n)$.

Proof. Repeated integration by parts gives $\widehat{f^{(k)}}(n) = (in)^k \hat f(n)$, so $|n^k \hat f(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely the series and all its term-by-term derivatives $\sum (ik)^j a(k) e^{ikx}$ converge uniformly by the M-test (as $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4); integrating the uniformly convergent series against e^{-inx} term by term gives $\hat f(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a, b) \to \mathbb{R}$ is analytic and $K \subset (a, b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \dots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k (x - c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k \ge n} \frac{k!}{(k-n)!} |a_k| \Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n} n! \sum_{j \ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n! \Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \dots$, then $c \le (R + 1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f : \mathbb{R} \to \mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty} |\hat f(n)|^{1/|n|} < 1$.

Proof. By Lemma 9.18 on $K = [-\pi, \pi]$, $\|f^{(k)}\|_{\sup} \le Ck!R^k$ for $k \ge 1$ (and enlarging C, also for $k = 0$). Then $|\hat f(n)| = |\widehat{f^{(k)}}(n)|/|n|^k \le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat f(n)| \le C(R + 1)e^{-|n|/(R+1)}$, and $\limsup |\hat f(n)|^{1/|n|} \le e^{-1/(R+1)} < 1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi}|f|<\infty$, then $\lim_{|n|\to\infty}\hat f(n)=0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i\sin nt)\,dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is odd, then the partial sums of $\sum_{n\ge1}\hat f(n)/n$ are bounded.

Proof. For odd f, $\hat f(n)=-\frac{i}{\pi}\int_0^\pi f(t)\sin nt\,dt$, so $\sum_{n\le N}\frac{\hat f(n)}{n}=-\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\le N}\frac{\sin nt}{n}\,dt$. The sums $\sum_{n\le N}\frac{\sin nt}{n}$ are bounded by $1+2\pi$ on $(0,\pi]$: the terms with $n\le 1/t$ are at most t each, and for $n>1/t$, summation by parts with (‡) of Chapter 6 gives at most $2\cdot t\cdot\pi/t$. Hence the partial sums are bounded by $\frac{1+2\pi}{\pi}\int_0^\pi|f|$. ∎

So $\sum_{n\ge2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f\in\mathcal{R}^1$, since $\sum\frac{1}{n\log n}=\infty$.

9.5 Selected exercises

Motivation. These problems from the textbook's exercise set explain why partial sums can misbehave while Cesàro means cannot, characterise the averaging methods that preserve limits, and show that bounded variation forces the Fourier coefficients to decay like $1/|n|$.

Exercise 9.5.3 (Lebesgue constants)

Show that $\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n(t)|\,dt\to\infty$ as $n\to\infty$; in fact it is at least $\frac{4}{\pi^2}(1+\frac12+\cdots+\frac1n)$.

Solution.

Step 1. D_n is even and $0<\sin\frac{t}{2}\le\frac{t}{2}$ on $(0,\pi]$, so

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n|=\frac{1}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{\sin\frac{t}{2}}\,dt\ge\frac{2}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{t}\,dt=\frac{2}{\pi}\int_0^{(n+\frac12)\pi}\frac{|\sin u|}{u}\,du,$$

by the substitution $u=(n+\frac12)t$.

Step 2. On $[k\pi,(k+1)\pi]$, $\frac1u\ge\frac{1}{(k+1)\pi}$ and $\int_{k\pi}^{(k+1)\pi}|\sin u|\,du=2$. Summing over $k=0,\dots,n-1$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n|\ge\frac{2}{\pi}\sum_{k=0}^{n-1}\frac{2}{(k+1)\pi}=\frac{4}{\pi^2}\sum_{k=1}^{n}\frac1k\to\infty.$$

By contrast the Fejér kernel satisfies $\frac{1}{2\pi}\int|K_n|=1$ because $K_n\ge0$; this is why Fejér's theorem holds for all continuous f while the partial sums may diverge (Example 9.2). ∎

Exercise 9.5.8 (Regular averaging methods)

Let $a_{mn} \geq 0$ $(m, n \geq 1)$ with $a_{mn} = 0$ for $n > m$ and $\sum_{n=1}^{m} a_{mn} = 1$ for every m. For a sequence (s_n) put $t_m = \sum_{n=1}^{m} a_{mn} s_n$. Show that the following are equivalent:
(a) for every convergent sequence, $\lim_n s_n = s$ implies $\lim_m t_m = s$;
(b) $\lim_{m\to\infty} a_{mn} = 0$ for each n.

Solution.

(a)⇒(b): Fix n_0 and let $s_{n_0} = 1$, $s_n = 0$ for $n \neq n_0$. Then $s_n \to 0$ and $t_m = a_{mn_0}$, so (a) gives $a_{mn_0} \to 0$.

(b)⇒(a):

Step 1. Since $\sum_n a_{mn} = 1$, $t_m - s = \sum_{n\leq m} a_{mn}(s_n - s)$.

Step 2. Given $\epsilon > 0$, choose N with $|s_n - s| < \epsilon$ for $n > N$. For $m > N$, using $a_{mn} \geq 0$,

$$|t_m - s| \leq \sum_{n\leq N} a_{mn}|s_n - s| + \epsilon \sum_{N<n\leq m} a_{mn} \leq \sum_{n\leq N} a_{mn}|s_n - s| + \epsilon.$$

Step 3. The first sum has N fixed terms, each tending to 0 as $m \to \infty$ by (b). Hence $\limsup_m |t_m - s| \leq \epsilon$ for every ϵ, i.e. $t_m \to s$.

Taking $a_{mn} = \frac{1}{m}$ for $n \leq m$ recovers Lemma 9.12. ■

Exercise 9.5.13 (Fourier coefficients of functions of bounded variation)

A function $f : [-\pi, \pi] \to \mathbb{C}$ is of bounded variation if $\sup_P \sum_i |f(x_i) - f(x_{i-1})| < \infty$ over all partitions P.
(a) Show that a function of bounded variation is a linear combination of increasing real functions.
(b) Show that for every increasing $f : [-\pi, \pi] \to \mathbb{R}$, $|\hat{f}(n)| \leq \frac{V_{-\pi}^{\pi}(f)}{\pi|n|}$ for $n = \pm 1, \pm 2, \ldots$.
(c) Show that the Fourier coefficients of a function f of bounded variation satisfy $|\hat{f}(n)| \leq \frac{C}{|n|}$ $(n \neq 0)$ for some constant $C > 0$.

Solution.

(a) Since $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$, the real functions $\operatorname{Re} f$ and $\operatorname{Im} f$ are of bounded variation. By the Jordan decomposition (Theorem 5.16 of Volume I), $\operatorname{Re} f = g_1 - h_1$ and $\operatorname{Im} f = g_2 - h_2$ with g_j, h_j increasing, so $f = g_1 - h_1 + i g_2 - i h_2$.

(b) Let $V = V_{-\pi}^{\pi}(f) = f(\pi) - f(-\pi)$ and $n \neq 0$.

Step 1. Choose $\theta \in \mathbb{R}$ with $e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt \geq 0$. Then

$$2\pi|\hat{f}(n)| = \operatorname{Re}\Big(e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt\Big) = \int_{-\pi}^{\pi} f(t) \cos(nt - \theta)\, dt.$$

Step 2. By the second mean value theorem (Corollary 5.25 of Volume I) there is $c \in [-\pi, \pi]$ with

$$\int_{-\pi}^{\pi} f(t)\cos(nt-\theta)\,dt = f(-\pi)\int_{-\pi}^{c} \cos(nt-\theta)\,dt + f(\pi)\int_{c}^{\pi} \cos(nt-\theta)\,dt.$$

Step 3. Since $\int_{-\pi}^{\pi} \cos(nt - \theta)\, dt = 0$, the right side equals $-V \int_{-\pi}^{c} \cos(nt - \theta)\, dt = -\frac{V}{n}(\sin(nc - \theta) - \sin(-n\pi - \theta))$, whose absolute value is at most $\frac{2V}{|n|}$.

Step 4. Hence $2\pi|\hat{f}(n)| \leq \frac{2V}{|n|}$, i.e. $|\hat{f}(n)| \leq \frac{V}{\pi|n|}$.

(c) Write $f = g_1 - h_1 + ig_2 - ih_2$ as in (a). By linearity of $f \mapsto \hat{f}(n)$ and (b),

$$|\hat{f}(n)| \le |\hat{g}_1(n)| + |\hat{h}_1(n)| + |\hat{g}_2(n)| + |\hat{h}_2(n)| \le \frac{C}{|n|},$$

with $C = \frac{1}{\pi}\big(V(g_1) + V(h_1) + V(g_2) + V(h_2)\big)$, where $V(\cdot) = V_{-\pi}^{\pi}(\cdot)$. For instance $f(x) = x$ has $|\hat{f}(n)| = \frac{1}{|n|}$ (Example 9.1), so the order $\frac{1}{|n|}$ cannot be improved. ■